# What gets scored—and what makes a passage context-dependent?

One actual English-web passage · these are single-passage scores · maximum 128 positions

A, B and C use the same trained language model and the same fixed canonical output head.

**The text is split once. The scored suffix stays identical in all three runs.**

**PREFIX · visible input · never scored**

```
.
1. Create following script, save it to your home folder as
"RunPreservingIconPos" and make it runnable.
NOTE! Rename all of these "icons.screen0-1280x1024.rc" filenames
in the script to correspond to your
```

**SUFFIX · all 32 tokens are scored**

```
 resolution in use!
cp -f ./icons.screen0-1280x1024.rc
./icons.
```

32 canonical tokens = 63 UTF-8 bytes

## 1 Decide whether the passage depends on its prefix

Use canonical input embeddings in both runs. Compare losses on the yellow suffix only.

**A · With the prefix**

BOS | Canonical prefix · 64 tokens | **SAME 32 suffix tokens · SCORED** → **0.765 BPB** (97 total positions)

**B · Remove the entire prefix**

BOS | **SAME 32 suffix tokens · SCORED** → **2.574 BPB** (33 total positions)

**Prefix benefit = B − A = 2.574 − 0.765 = 1.809 BPB**

1.809 > 0.555 threshold → INCLUDE this passage in the context-dependent subset.

The 0.555 threshold was fixed from discovery English-web passages (75th percentile of prefix benefit).

## 2 Separately measure this tokenizer's transfer penalty

Use the same prefix text, encoded by tokenizer #0771. Keep the suffix canonical.

**C · Transferred input embeddings**

BOS | Same prefix text · 40 native tokens | **SAME 32 suffix tokens · SCORED** → **1.186 BPB** (73 total positions)

**Transfer gap = C − A = 1.186 − 0.765 = +0.421 BPB**

C is used to measure transfer loss. It is NOT used to decide whether the text is context-dependent.

**How scoring works inside every yellow box**

Predict token 1 → supply the correct token 1 → predict token 2 → … → predict token 32.

Each prediction sees earlier correct suffix tokens, even in B. Future suffix tokens are masked.

Sum all 32 prediction losses in bits, then divide by the same 63 bytes. BOS and prefix add no loss.

For the table: all passages → natural-text column; passages passing step 1 → context-dependent column.

Compute each tokenizer's average gap on that set, then take the median across the 12 tokenizers in its row.